

Task 10: Price Category Prediction and Evaluation

Mobile Price Prediction Classification Project | 20 September 2026

1. Business Requirement

The mobile-phone company requires a machine-learning solution that predicts the appropriate price category of a phone from technical specifications such as RAM, internal storage, battery capacity, camera quality, screen resolution, processor-related features, connectivity, and other hardware characteristics.

2. Task Objective

Task 10 generates price-category predictions for unseen mobile phones and evaluates the Random Forest classifier trained in Task 9. Accuracy, precision, recall, F1-score, and a confusion matrix are used to measure classification quality across the three business price categories.

3. Evaluation Methodology

The saved Random Forest model was loaded from random_forest_model.pkl and applied to the 204 records in testing_dataset.csv. The target column was Price_Category_Encoded, with class 0 representing Below 15,000, class 1 representing 15,000-29,999, and class 2 representing 30,000 and above. The testing records were not used during model training.

4. Evaluation Summary

Measure	Result
Testing records	204
Predictor features	26
Target column	Price_Category_Encoded
Price categories	3
Evaluation type	Held-out test evaluation

5. Overall Performance

Metric	Value
Accuracy	85.29%
Weighted precision	85.38%
Weighted recall	85.29%
Weighted F1-score	85.32%

The model correctly classified 85.29% of the unseen mobile phones. The close values for weighted precision, recall, and F1-score indicate balanced performance across the test set, while the weighted scores account for the number of records in each price category.

6. Class-Wise Performance

Price category	Precision	Recall	F1-score	Support
Below 15,000	89.83%	91.38%	90.60%	58
15,000-29,999	78.95%	81.08%	80.00%	74
30,000 and above	88.41%	84.72%	86.52%	72

7. Confusion Matrix

Actual \ Predicted	Below 15,000	15,000-29,999	30,000 and above
Below 15,000	53	5	0
15,000-29,999	6	60	8
30,000 and above	0	11	61

The confusion matrix shows 53, 60, and 61 correct predictions for the low, middle, and high price categories respectively. Most errors occur between neighboring price categories, especially between the middle and high categories, which is reasonable because their technical specifications can overlap.

8. Deliverables

Implementation: price_category_prediction_evaluation.py Predictions: price_category_predictions.csv Evaluation results: price_category_evaluation_results.json Confusion matrix: price_category_confusion_matrix.csv

9. Conclusion

Task 10 is complete. The Random Forest model successfully generated price-category predictions for unseen mobile phones and achieved 85.29% test accuracy. The evaluation outputs provide the evidence required to assess model quality and support the business goal of automated mobile-phone price classification.

Generated from price_category_evaluation_results.json by generate_task_10_report.py.